Zafar Hussain

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Summary

Postdoctoral Researcher at Aarhus University's Center for Humanities Computing, specializing in Large Language Models (LLMs). My research focuses on designing Retrieval-Augmented Generation (RAG) systems. I am part of the team building Lex AI for the Danish national encyclopedia, working across retrieval, generation, evaluation, and production deployment.

Professional Experience

Postdoctoral Researcher, Aarhus University, Denmark *Sep 2025 – Present*

- Building Lex AI, an LLM-based service for the Danish national encyclopedia (Lex.dk).
- Developing and evaluating the end-to-end RAG pipeline that powers the service.
- Researching conformity and human-like social behavior of LLMs in social psychology experiments.

Doctoral Researcher, University of Helsinki, Finland *Jan 2021 – Aug 2025*

- Developed machine learning models that learn the syntax and structure of semi-natural language (command-line commands) to support the detection of malicious activity.
- Contributed to the IVVES, IML4E, and ELFMO projects, joint research collaborations between the University of Helsinki and industry partners.

Research Assistant, University of Helsinki, Finland *Apr 2019 – Dec 2020*

- Studied tools and methods for testing machine learning systems.
- Analyzed heterogeneous Finnish newspaper data using statistical methods to study the development of public discourse, and built ML models to predict newspaper lifespans and forecast feature changes.

Research Assistant, University of Oulu, Finland *Jan 2016 – Oct 2016*

- Evaluated wake-up-radio-based MAC and physical layers for energy-efficient wireless body area networks.

Software QA Engineer, Systems Limited, Pakistan *Jan 2013 – Aug 2014*

- Delivered stable releases of apparel and e-commerce projects through manual and automated testing.

Software QA Engineer, NextBridge (Pvt) Ltd, Pakistan *May 2011 – Dec 2012*

- Managed automated testing for software and large database applications using Selenium and QC.

Education

Ph.D. in Computer Science, University of Helsinki, Finland *2021 – 2025*
Thesis: Learning Command Syntax and Detecting Similarities for Enhanced Cybersecurity through Data Analysis

M.Sc. in Data Science, University of Helsinki, Finland *2018 – 2020*

M.Sc. in Wireless Communications Engineering, University of Oulu, Finland *2014 – 2016*

B.Sc. in Computer Science, Government College University, Pakistan *2006 – 2010*

Skills

- **Research Areas:** NLP, LLMs, Retrieval-Augmented Generation, Deep Learning, LLM Evaluation
- **Programming:** Python, R, MATLAB, SQL/MySQL, MongoDB
- **ML Frameworks & Libraries:** PyTorch, TensorFlow, scikit-learn, Pandas, NumPy, NetworkX
- **Tools & Cloud:** AWS, Tableau, Selenium
- **Certification:** ISTQB Certified Tester, Foundation Level (FI-CTFL-2607), Finnish Software Testing Board, 2016

SELECTED PUBLICATIONS

1. Zafar Hussain, Kristoffer L. Nielbo. **Conformity Mitigations in Large Language Models Lie on a Single Resistance-Receptivity Frontier.** *arXiv preprint arXiv:2608.11247*, 2026.
2. Zafar Hussain, Kristoffer L. Nielbo. **The Coverage Illusion: From Pre-retrieval Routing Failure to Post-retrieval Cascades in a Production RAG System.** *Proceedings of the 1st Workshop on Grounding Language Models (GroundLM 2026)*, pp. 937–953, Budapest, Hungary. ACL, 2026.
3. Zafar Hussain, Jukka K. Nurminen, Perttu Ranta-aho. **Training a Language Model to Learn the Syntax of Commands.** *Array*, vol. 23. Elsevier, 2024.
4. Zafar Hussain, Lalli Myllyaho, Jukka K. Nurminen. **Learning the Structure of Commands by Retraining a Language Model.** *Proceedings of the 9th International Conference on Machine Learning Technologies (ICMLT 2024)*, Oslo, Norway. ACM, 2024.
5. Zafar Hussain, Jukka K. Nurminen, Perttu Ranta-aho. **Learning the Structure of Commands by Detecting Random Tokens Using Markov Model.** *Proceedings of the 8th International Conference on Machine Learning Technologies (ICMLT 2023)*, Stockholm, Sweden. ACM, 2023.
6. Zafar Hussain, Jukka K. Nurminen, Tommi Mikkonen, Marcin Kowiel. **Combining Rule-Based System and Machine Learning to Classify Semi-Natural Language Data.** *Intelligent Systems and Applications (IntelliSys 2022)*, LNNS vol. 542. Springer, 2022.
7. Zafar Hussain, Jukka K. Nurminen, Tommi Mikkonen, Marcin Kowiel. **Command Similarity Measurement Using NLP.** *10th Symposium on Languages, Applications and Technologies (SLATE 2021)*, OASIcs vol. 94. Schloss Dagstuhl, 2021.